TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

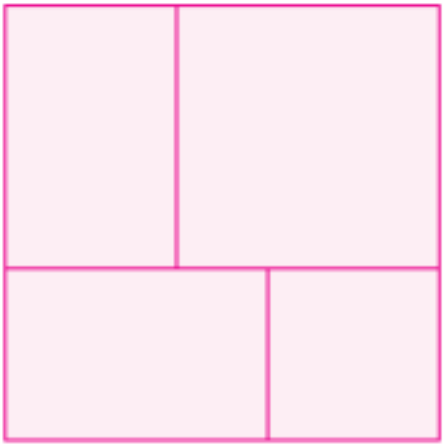
TIMO

GRADE 8

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

1.



In the figure, two different squares and two identical rectangles are drawn inside the large square with area of 225 cm². The long side of one of the rectangles is 3 cm more than the short side.

According to this, what is the difference in the areas of the two squares drawn inside?

A) 54 B) 49 C) 45 D) 42 E) 36

2. The following ratios are given among the numbers a, x, y, and z.

$$\frac{x}{y}=\frac{3}{4} \qquad \frac{x}{z}=\frac{1}{2} \qquad \frac{a}{z}=\frac{5}{6}$$

What is the value of the ratio of $\frac{z^2-y^2}{a\cdot x}$?

A) $\frac{3}{4}$ B) $\frac{2}{5}$ C) 1 D) $\frac{4}{3}$ E) $\frac{5}{2}$

3. In the figure the triangle KLM whose vertices lie on the sides of rectangle ABCD is given.
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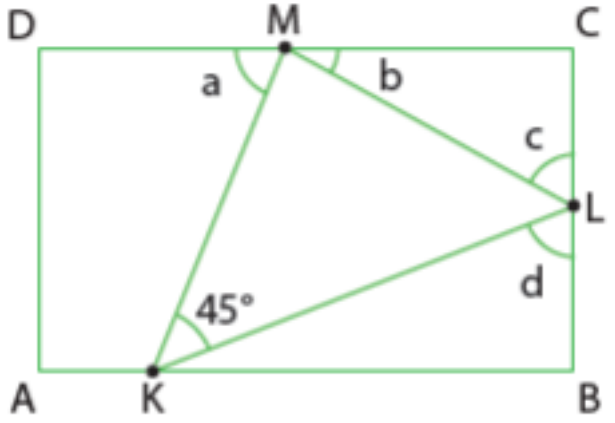


If $m(\widehat{MKL}) = 45^o$
then what is the value of the sum a+b+c+d in degrees?

A) 135 B) 180 C) 195 D) 210 E) 225

4. Each of the different numbers obtained by writing each digit as much as itself will be written side by side and summed as shown below.

1+22+333+4444+…+999999999

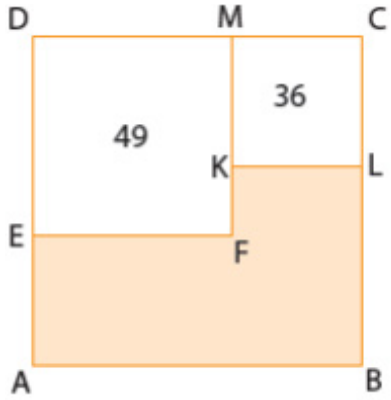
What is the sum of the last two digits of the number obtained by this summation?

A) 13 B) 14 C) 15 D) 16 E) 17

5. Find the value of

$$\frac{1012^2 + 2024^2 + 3036^2 + 4048^2}{1012 + 2024 + 3036 + 4048}$$

A) 1012 B) 2024 C) 2530 D) 3036 E) 4048

6. 

There are 3 squares in the figure above.
Since the area of the smallest square is 36 cm² and the area of the median square is 49 cm², what is the perimeter of the shaded shape?

A) 35 B) 37 C) 40 D) 43 E) 48

7. a, b and c are reel numbers satisfying the system

$$\frac{1}{a+b}+\frac{1}{a+c}+\frac{1}{b+c}=3$$

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}=18$$

Find a+b+c?

A) 5 B) 6 C) 7 D) 8 E) 9

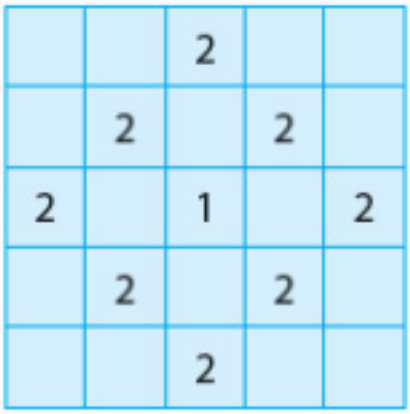
8. Ahmet and Burak are swimming in a river which is 700 metres long and flows at a constant speed. The swimming speeds of each of Ahmet and Burak are different from each other and the speed of each in the same direction with the current is 2 times their speed against the current. When they start swimming against the current at the same time, Burak finishes 700 metres and when he turns back, he meets Ahmet after swimming 70 metres.
Accordingly, when Burak finishes the race, how many more metres does Ahmet have to swim to finish the race?

A) 350 B) 320 C) 300 D) 280 E) 240

9. In a 5 × 5 grid, a frog that starts moving from the desired square has two different movement abilities.

- It can jump over a square in the same row or column and move to the 2nd square.
- It can move to the neighbouring square on the diagonal.

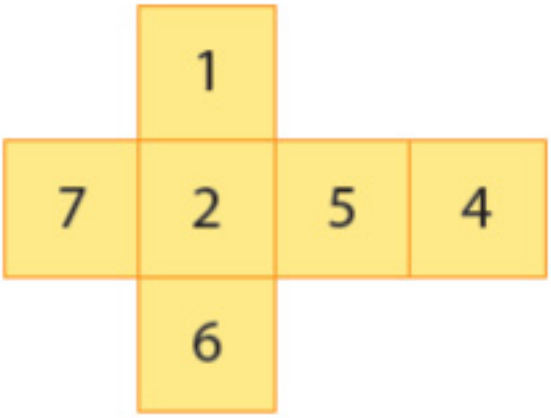
For example in the figure below the frog in square 1 can move to one of the squares labelled 2.



If the frog can go to each small square at most once, at most how many squares can it go?

A) 12 B) 13 C) 14 D) 24 E) 25

10. The open form of a cube with numbers written on it is given below. When this cube is closed, three faces meet at each corner.



According to this, what is the maximum sum of the numbers written on the three faces that meet at any corner?

A) 14 B) 15 C) 16 D) 17 E) 18

11. In a class of 30 students, 3 different books will be distributed to each student. These books belong to maths, physics, chemistry and biology courses. The books of a course are identical in themselves .Since a total of 25, 24 and 21 maths, physics and chemistry books were distributed respectively, how many students did not receive a biology book?

A) 9 B) 10 C) 11 D) 12 E) 13

12. The following numbers are written according to a rule.

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

16 ...

⋮

According to this rule what is the smallest number in the row which includes 200?

A) 197 B) 191 C) 188 D) 183 E) 171

13. The following number is obtained by writing consecutive integers from 1 to 50 side by side.
1234…4950
Deleting some digits from the number above, it is desired to obtain the largest number with 81 digits. What is the sum of the deleted digits?

A) 46 B) 45 C) 37 D) 36 E) 32

14. A 2024 digits natural number which has all its digit 2 is given.
What is the sum of the digits of the number obtained by multiplying this number by 2222?

A) 14163 B) 14170 C) 14177 D) 14184 E) 14191

15. Efe, who goes for a ride with his bicycle in the park, is at the entrance of the park at 08.00 am. There is one of each kind of tree in this park. Efe passes by acacia tree at 08.20 am, plane tree at 08.35 am, poplar tree at 09.05 am, cedar tree at 09.45 am, hornbeam tree at 10.45 am and reaches the exit of the park at 11.00 am. After resting for half an hour, it sets off towards the entrance from the same road it came from, at a constant speed of half the speed of its arrival.
According to this, at what time does he pass by the plane tree on the way back?

A) 16.40 B) 16.20 C) 16.00 D) 15.50 E) 15.40

16. While Hlya, Hatice and Bilge were studying together, they took a mock test and timed it and stayed in the test for 61, 65 and 53 minutes, respectively.
Since Hatice started the test at the same time as Hülya and finished the test at the same time as Bilge, how many minutes did the three of them take the test at the same time?

A) 41 B) 43 C) 45 D) 47 E) 49

17. 25 students are lined up in a row, one after the other. Each student is either a good student who always tells the truth or a mischievous student who always lies. Omar, a good student, is the 13th person in this line. Everyone except Omar says, “There are exactly 6 mischief makers between me and Omar”.
According to this how many good students are there in this line?

A) 6 B) 7 C) 12 D) 13 E) 14

18. A number divisible by 12 consists of only 0s and 1s.
What is the number of the digits of the smallest number that satisfies this condition?

A) 3 B) 4 C) 5 D) 6 E) 8

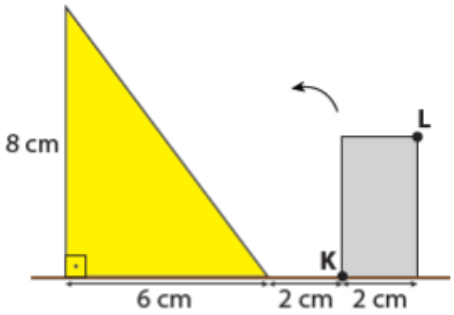
19. The rule of the number sequence whose first 4 terms are given below is as follows:
 From the 5th term onwards, each term is the largest digit different from the previous 4 terms.
9, 7, 6, 2, ...
According to this, what is the 2024th term of this sequence?

A) 8 B) 9 C) 7 D) 6 E) 5

20. Each side of a cube will be coloured red or black provided that each of the surface of the cube will have at least one black partition.
According to this, what is the minimum number of separations that can be coloured black?

A) 2 B) 3 C) 4 D) 5 E) 6

21.



The right side lengths of the yellow right triangle in the figure are 6 cm and 8 cm. The short side of the grey rectangle drawn on the same ground as the triangle is 2 cm. When the rectangle is rotated 90° around point K in the direction of the arrow, point L of the rectangle lies on the hypotenuse of the triangle.
Since the distance between the triangle and the rectangle is 2 cm, what is the area of the region where the triangle and the rectangle intersect after the rectangle is rotated in cm^2?

A) $\frac{3}{2}$ B) $\frac{3}{4}$ C) $\frac{5}{2}$ D) 2 E) $\frac{7}{4}$

22. If $a+b=3$ and $a^2+b^2=7$, find the value of a^5+b^5 ?

A) 75 B) 92 C) 105 D) 123 E) 136

23. How many integer solutions does the equation $x^2 - y^2 = 202$ have?

A) 0 B) 1 C) 2 D) 4 E) 101

24. How many 7-digit numbers whose sum of the digits is 3 are there?

A) 18 B) 24 C) 28 D) 30 E) 32

25. Given x, y, z $\in \{0, 1, 2\}$ and a, b, c $\in Z^+$ If

$$x \cdot 3^a + y \cdot 3^b + z \cdot 3^c = 423$$

then what is the value of

$$\frac{a+b+c}{x+y+z} \ ?$$

A) $\frac{17}{5}$ B) $\frac{13}{6}$ C) $\frac{13}{5}$ D) $\frac{17}{6}$ E) $\frac{11}{5}$